

3(a)(i)	<i>any two from:</i> 1 group of (morphologically) similar organisms ; 2 that interbreed to produce fertile offspring ; 3 occupy same niche ; 4 AVP ; e.g. detail of molecular similarity	2
---------	---	----------

Question	Answer	Marks
3(a)(ii)	<i>any five from:</i> 1 <i>ref. to</i> geographical isolation ; 2 no, gene flow / interbreeding, between, A and B / populations ; 3 different, selection pressures / environmental conditions (for A and B populations) ; 4 different mutations occur (in A and B populations) ; 5 some mutations make individuals better adapted ; 6 those individuals, survive / reproduce ; 7 pass on advantageous alleles ; 8 <i>ref. to</i> many generations ; 9 reproductive isolation ; 10 allopatric speciation ;	5
3(b)(i)	<i>any two from mp1-4 and one for mp5</i> <i>any two from:</i> 1 DNA sample placed in well (at end of gel) ; 2 <i>ref. to</i> electric field / current ; 3 (negatively charged) DNA attracted to, anode / positive electrode ; 4 shorter fragments move further in unit time / ora ; <i>one from:</i> 5 compare band position to identify cat species ;	3

Question	Answer	Marks
3(b)(ii)	any two from: 1 quicker test ; 2 cheaper / good for low income countries ; 3 portable / no special lab needed / no need to send off sample to be tested ; 4 many sequences can be stored / <i>ref. to</i> bioinformatics ; 5 AVP ; e.g. more successful prosecutions deters others form illegal trading / new enforcement measures available	2

Question	Answer	Marks
----------	--------	-------